

IY-481: Project Deliverable 1

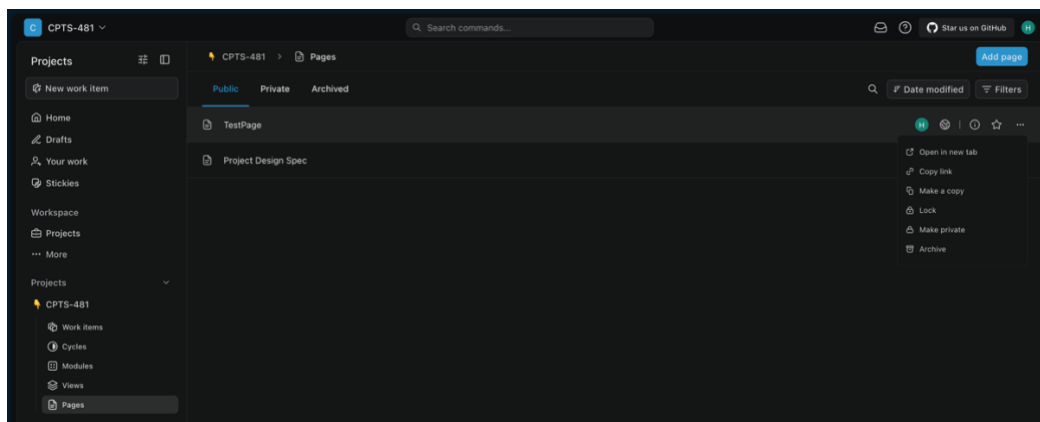
WSU Repository: <https://github.com/WSU-CptS-481-Spring-2026/IY-481-Plane>

Team:

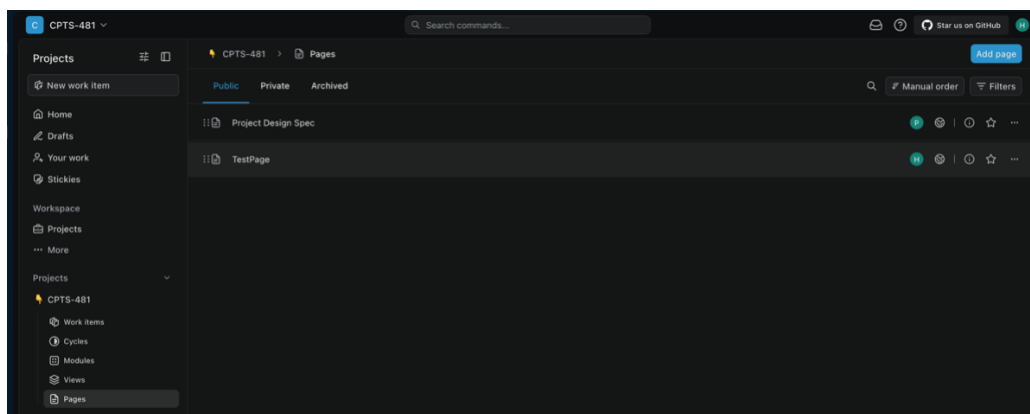
- Hiruna Yevin Dissanayake
- Ivan Quintero

Individual contributions:

- Hiruna Yevin Dissanayake:
 1. #4443: Ability to order pages (Feature)



(Before Feature implementation)



(After feature implementation)

- Files altered:
 - apps/api/plane/app/serializers/page.py:
 - Calls sort-order
 - apps/api/plane/app/views/page/base.py:

- Sets default order
- apps/web/core/components/pages/list/block.tsx:
 - Imports drag handles
 - Implements reordering logic
- apps/api/plane/tests/contract/app/test_pages_app.py (Test file)
 - test_list_pages_sorted_by_sort_order_desc
 - Calls pages
 - Verifies returns 200 for OK status
 - Checks that pages are sorted by sort_order
 - Checks each row has a sort_order field
 - test_patch_sort_order_updates_page
 - Sends patch to update pages from sort order 150 to 250
 - Check change in value
 - test_patch_sort_order_allowed_when_page_is_locked
 - Marks all low pages as locked
 - Check that locked pages can still be sorted
 - test_patch_non_sort_field_blocked_when_page_is_locked
 - Mark High pages as locked
 - Try to change name or locked page after sorting
 - Check for 400 BAD REQUEST due to lock
 - Verify name has not changed

```
(base) user@Yevins-MacBook-Pro IY-481-Plane % docker compose -f docker-compose-local.yml exec api python -m pytest plane/tests/contract/app/test_pages_app.py -q
===== test session starts =====
platform linux -- Python 3.12.5, pytest-7.4.0, pluggy-1.6.0
django: settings: plane.settings.test (from ini)
rootdir: /code
configfile: pytest.ini
plugins: cov-4.1.0, django-4.5.2, xdist-3.3.1, mock-3.11.1, Faker-25.0.0, anyio-4.12.1
collected 4 items

plane/tests/contract/app/test_pages_app.py ....
===== 4 passed in 1.74s =====
(base) user@Yevins-MacBook-Pro IY-481-Plane %
```

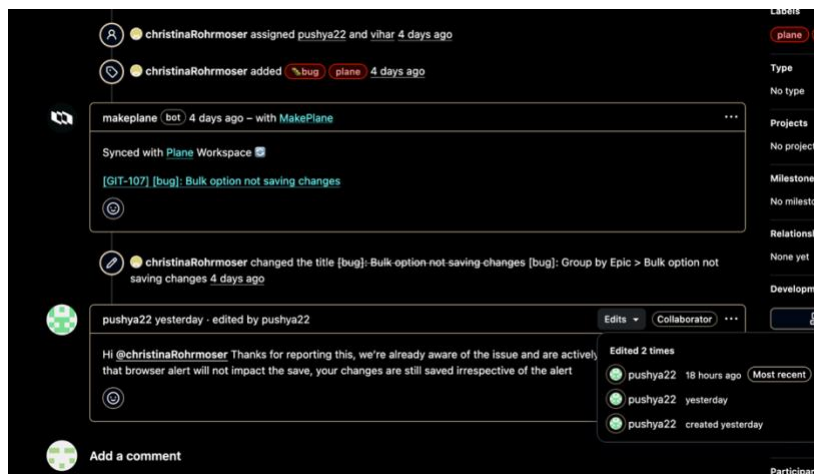
(All tests passing on pytest)

2. #7550: Work Item Opened from Home Page Recents Missing Cycle and Module (Simple Bug)
 - Between implementing tests to catch bug and implementing fix this issue was fixed without issue being marked as closed or being connected to a specific commit.

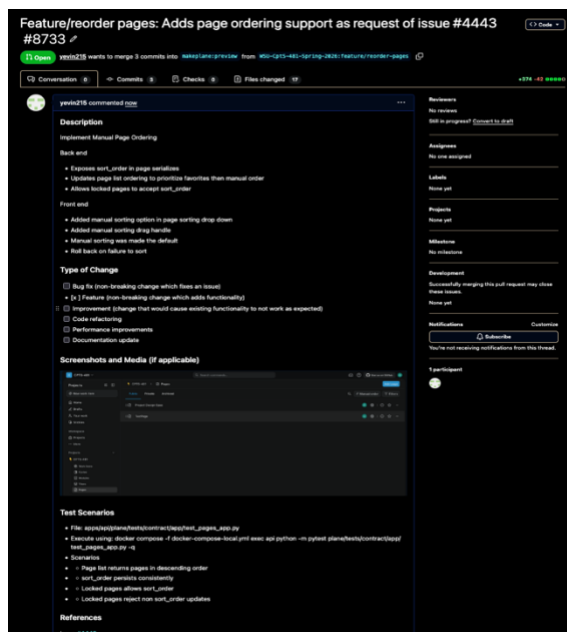
- Contacted maintainers about the issue being fixed.



3. #8683 - Group by Epic > Bulk option not saving changes (Advanced Bug)
 - While implementing the fix for this bug the dev team took over the fix for this 1 day before fix could be implemented completely.



4. Pull Requests



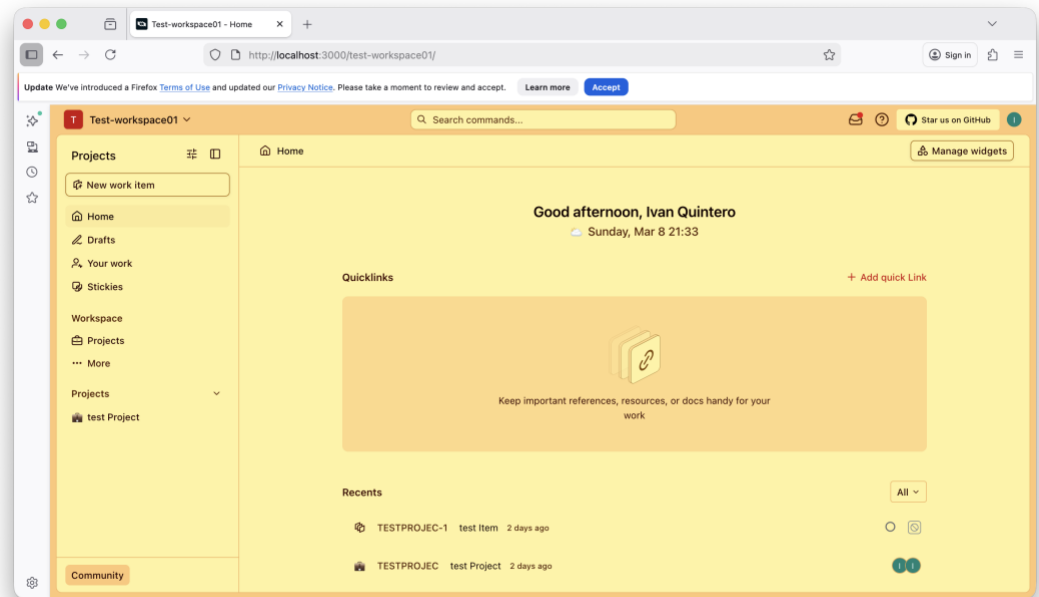
NOTE:

Pull requests for issues 7550 and 8683 were not created since they were either resolved or taken over by the dev team of the application.

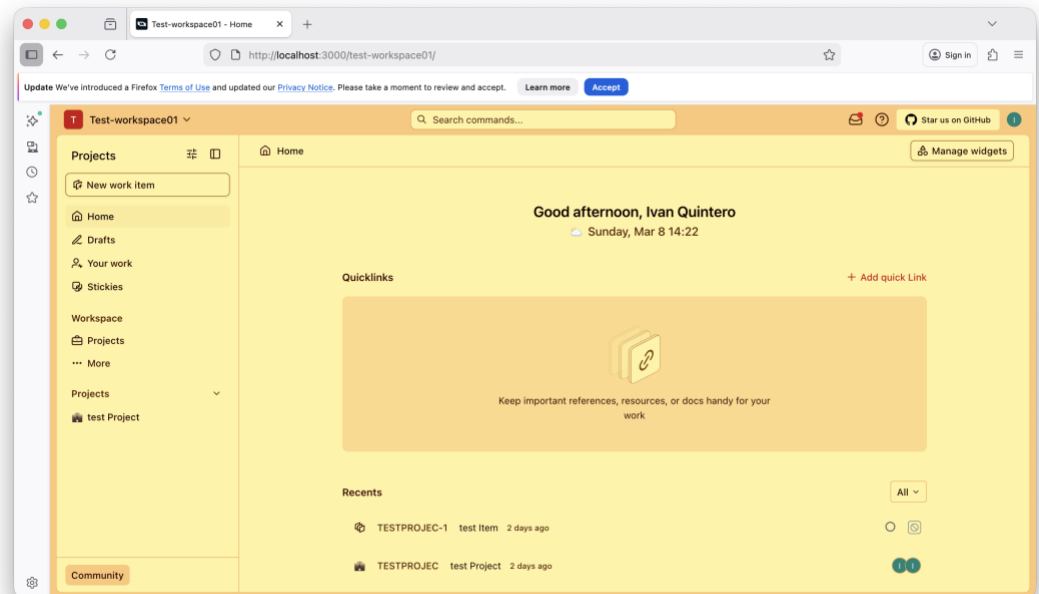
Therefore, after implementing issue 4443 I focused on the creation of the report heavily freeing up Ivan to finish his change requests.

This was needed due to the absence of participation from our 3rd teammate Ryan who we were unable to reach.

- Ivan Quintero:
 1. #8580 – Greeting Message (Bug)
 - **Before**



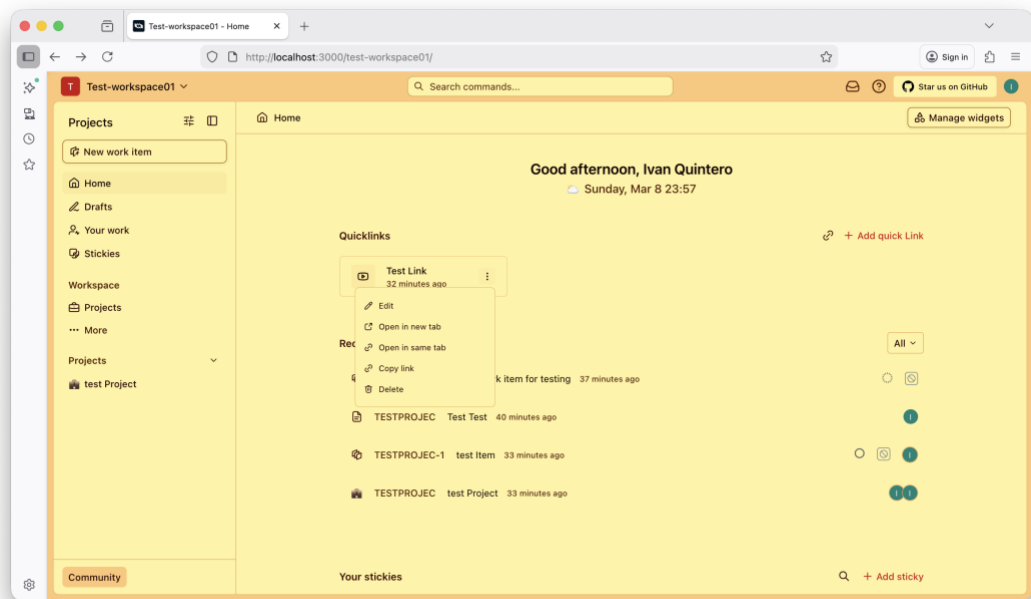
- **After**



- Files altered:

- apps/web/core/components/home/user-greetings.tsx
 - Removed an unused import to fix linting error.
 - Added logic to pick the right timezone.
 - Reads the profile timezone, treats values like UTC, Etc/UTC, or any variance of UTC, or missing/empty as “unset/default”. If it is unset/default, it falls back to browser/OS timezone, otherwise it uses the profile timezone.

2. #8621 – Option to open quicklinks in the same page (Feature)



- Files altered:
 - apps/web/core/components/home/widgets/links/contants.ts
 - Defined two constants to be used by the quicklinks widget.
 - packages/il8n/src/locales/en/translations.ts
 - Added 3 new translation strings under the Quicklinks section
 - apps/web/core/components/home/widgets/links/link-detail.tsx
 - Added a localStorage-backed setting to decide whether quicklinks should open in the same tab.
 - Added the “open in same tab” behavior
 - Changed the primary click behavior
 - Updated memo dependency array
 - apps/web/core/components/home/widgets/links/root.tsx
 - Added new imports

- Added a new state (openInSameTab)
- Added new toggle handler to flip the preference for links (to open in new tab or same)
- UI changes for the toggle button
- apps/api/plane/tests/contract/app/test_quick_links.py
 - Tests various CRUD functions of the Quicklinks home widget

```
api — zsh — 80x24

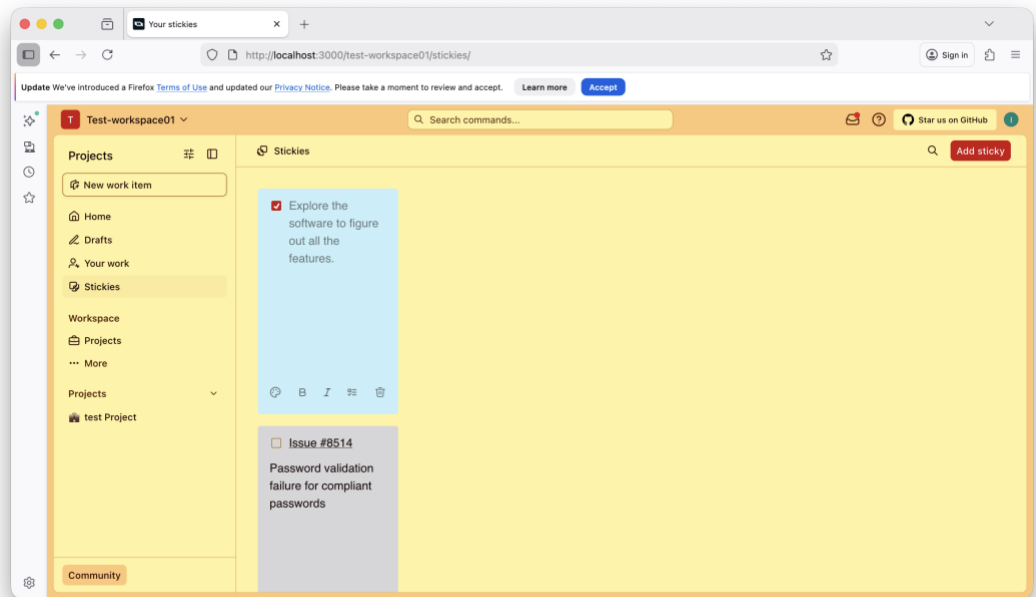
collected 9 items

plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_list_empty
Using existing test database for alias 'default' ('test_plane_test')...
PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_create_quick_link_success PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_create_quick_link_url_only PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_create_quick_link_invalid_url PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_create_quick_link_empty_data PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_retrieve_quick_link PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_partial_update_quick_link PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_destroy_quick_link PASSED
plane/tests/contract/app/test_quick_links.py::TestQuickLinksAPI::test_retrieve_nonexistent_returns_404 PASSED

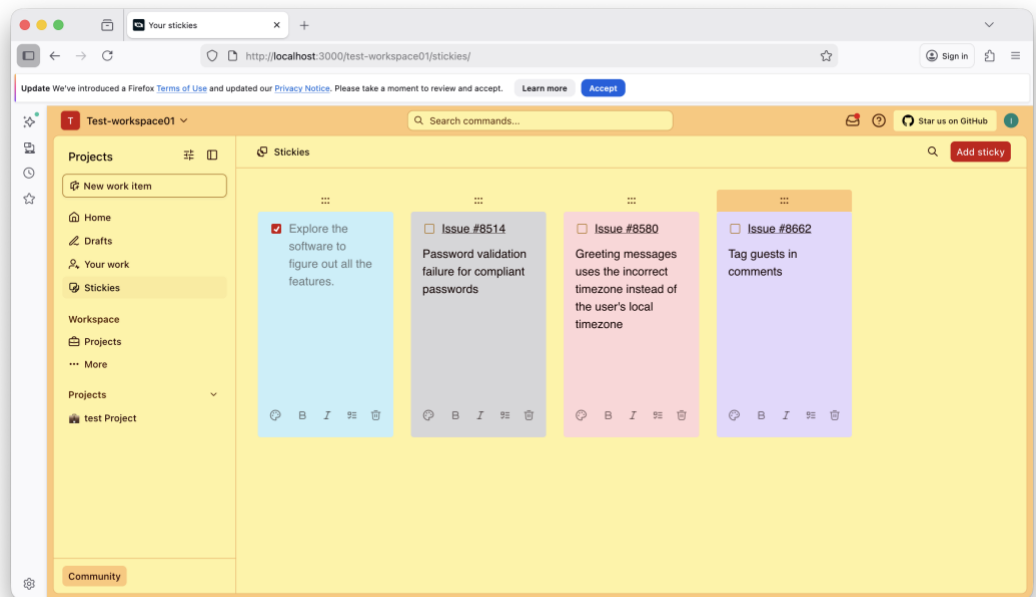
===== 9 passed in 2.75s =====
(.venv) (base) ivanquintero@Ivans-MacBook-Air api %
```

3. #8488 – Stickies reordering (Feature)

- **Before**



- **After**



- **Files altered:**

- apps/web/core/store/sticky/sticky.store.ts
 - Guard clauses to prevent trying to “move” a sticky onto itself.
- apps/web/core/components/stickies/layout/stickies-list.tsx

- Removed Masonry layout, switched from Masonry to CSS Grid layout
- Changed how itemWidth is handled. It is now hard-set to 100% since we no longer use Masonry.
- Refactored and moved getColumnCount. It is now a helper function.
- apps/web/core/components/stickies/layout/sticky-dnd-wrapper.tsx
 - Added a drag-handle since it was set to the entire sticky before and was not working.
 - Variable naming was cleaned up to fix linting errors
 - Added more props (inFirstRow)
- apps/api/plane/api/views/sticky.py
 - Changed the order of the stickies to be ordered by highest sort_order first, and then by newest first as tie-breaker.
- apps/api/plane/tests/contract/api/test_sticky.py
 - test_default_sort_order_assignment
 - Creates a sticky with only workspace and name. Asserts sort_order is 65535.
 - Creates another sticky in the same workspace. Asserts its sort_order is 132070, i.e the next slot.
 - test_api_returns_ordered_list
 - Creates two stickies with explicit sort_order: one “Last” with 200, one “First” with 100.
 - Calls GET /api/workspaces/{workspace_slug}/stickies/ with the Django test client.
 - Asserts the first item in the response is “First” and the second is “Last”.

4. Pull Requests

[Bug] - Greeting message uses incorrect timezone (Issue #8580)

#8735 

 Open [Ivanq50](#) wants to merge 2 commits into [makeplane:preview](#) from [WSU-CptS-481-Spring-2026:fix-8580-greeting-message](#) 

 Conversation 14

 Commits 2

 Checks 0

 Files changed 5



[Ivanq50](#) commented 19 minutes ago • edited ▾

...

Description

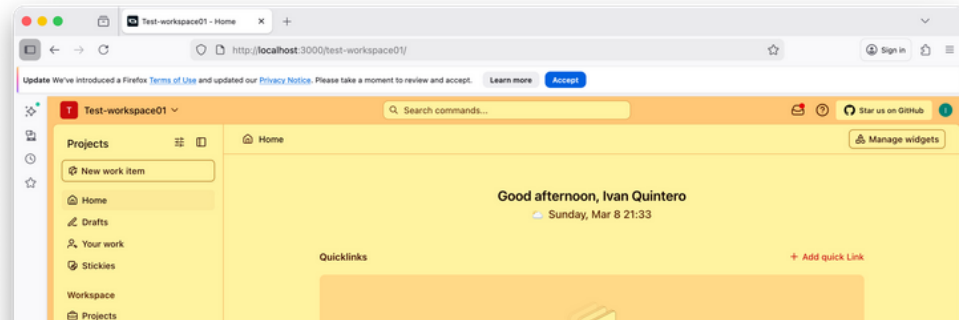
Added logic to select the correct timezone for the user instead of defaulting to UTC. Reads the profile timezone and treats values like UTC, etc/UTC, any variance of UTC, or missing/empty values as "unset/default". If it is unset/default, it falls back to browser/OS timezone, otherwise, it uses the profile timezone.

Type of Change

- ☒ Bug fix (non-breaking change which fixes an issue)
- ☐ Feature (non-breaking change which adds functionality)
- ☐ Improvement (change that would cause existing functionality to not work as expected)
- ☐ Code refactoring
- ☐ Performance improvements
- ☐ Documentation update

Screenshots and Media (if applicable)

Correct time: 2:33PM PST



[Feature] - Adding Stickies reordering feature (Issue #8488) #8736

 Open [Ivanq50](#) wants to merge 5 commits into [makeplane:preview](#) from [WSU-Cpt5-481-Spring-2026:fix-8488-sticky-reorder](#) 

 Conversation 1  Commits 5  Checks 0  Files changed 22



Ivanq50 commented 3 minutes ago • edited by coderabbitai 

Description

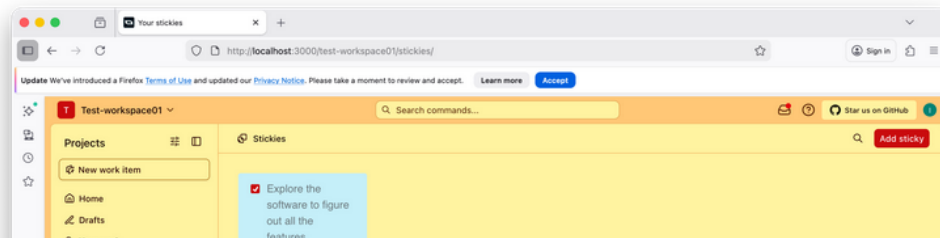
Implemented the requested feature to allow the user to customize the order of the stickies. This feature only works on the Stickies tab (not on the home page via widget). Added a drag-handle to the Stickies to make it clear for the user where to select and drag. Changed the format from Masonry to CSS Grid as this fit the Stickies better.

Type of Change

- ☐ Bug fix (non-breaking change which fixes an issue)
- ☒ Feature (non-breaking change which adds functionality)
- ☐ Improvement (change that would cause existing functionality to not work as expected)
- ☐ Code refactoring
- ☐ Performance improvements
- ☐ Documentation update

Screenshots and Media (if applicable)

Before



[Feature]- Option to open Quicklinks in same page (Issue #8621) #8734

Open

Ivanq50 wants to merge 5 commits into [makeplane:preview](#) from [WSU-CptS-481-Spring-2026:fix-8621-quicklinks](#)

Conversation 13

Commits 5

Checks 0

Files changed 22



Ivanq50 commented 51 minutes ago • edited by coderabbitai bot

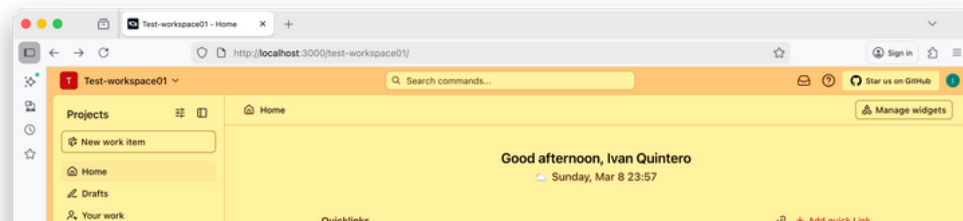
Description

Added an option for the user to open links within the same page if desired. Placed a toggle button to set that preference, but the user also has the option to select that preference individually by clicking the "Open in new tab" or "Open in same tab" option after clicking the more options menu next to the link.

Type of Change

- ☐ Bug fix (non-breaking change which fixes an issue)
- ☒ Feature (non-breaking change which adds functionality)
- ☐ Improvement (change that would cause existing functionality to not work as expected)
- ☐ Code refactoring
- ☐ Performance improvements
- ☐ Documentation update

Screenshots and Media (if applicable)



Pre Factoring Activities

1. Repository Examination: Before implementing changes, our team examined the following repositories thoroughly to gain a thorough program comprehension and code quality assessment. Helping us understand where specific features and changes should occur.
 - apps/api/plane/app/serializers
 - apps/api/plane/app/views/page
 - apps/api/plane/api/views
 - apps/web/core/components
 - apps/web/core/store
 - packages/types

- packages/utils

2. Code Search: Our team used ripgrep for code search, helping us find exactly where certain phrases may be located to help up located exact features

```
(base) user@Yevins-MacBook-Pro IV-481-Plane % rg "page" apps
apps/admin/store/workspace.store.ts
111:  * @description Fetches the next page of workspaces
115:  if (!this.paginationInfo || this.paginationInfo.next_page_results === false) return [];

apps/space/store/helpers/filter.helpers.ts
26:  // if cursor exists, use the cursor. If it doesn't exist construct the cursor based on per page count
27:  const pageCursor = cursor ? cursor : groupId ? `${options.perPageCount}:1:0` : `${options.perPageCount}:0:0`;
32:  cursor: pageCursor,
33:  per_page: options.perPageCount.toString(),

apps/api/templates/csrf_failure.html
59:  <li>Refresh the page and try submitting the form again.</li>
```

(Example use of ripgrep)

3. Static Code Analysis: Before implementing any major changes, our team performed static code analysis using Pylint on files we would have to modify to identify potential maintainability issues and code smells.

```
(base) user@Yevins-MacBook-Pro IV-481-Plane % pylint apps/api/plane/app/serializers/page.py
***** Module plane.app.serializers.page
apps/api/plane/app/serializers/page.py:1:0: C0114: Missing module docstring (missing-module-docstring)
apps/api/plane/app/serializers/page.py:26:0: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:36:4: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:36:4: R0903: Too few public methods (0/2) (too-few-public-methods)
apps/api/plane/app/serializers/page.py:130:0: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:133:4: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:133:4: R0903: Too few public methods (0/2) (too-few-public-methods)
apps/api/plane/app/serializers/page.py:137:0: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:138:4: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:138:4: R0903: Too few public methods (0/2) (too-few-public-methods)
apps/api/plane/app/serializers/page.py:154:0: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:155:4: C0115: Missing class docstring (missing-class-docstring)
apps/api/plane/app/serializers/page.py:155:4: R0903: Too few public methods (0/2) (too-few-public-methods)
apps/api/plane/app/serializers/page.py:174:0: W0223: Method 'create' is abstract in class 'BaseSerializer' but is not overridden in child class 'PageBinaryUpdateSerializer' (abstract-method)
apps/api/plane/app/serializers/page.py:196:8: C0103: Variable name 'e' doesn't conform to snake_case naming style (invalid-name)
apps/api/plane/app/serializers/page.py:7:0: C0411: standard import 'import base64' should be placed before 'from rest_framework import serializers' (wrong-import-order)
apps/api/plane/app/serializers/page.py:11:0: C0411: third party import 'from plane.utils.content_validator import validate_binary_data, validate_html_content' should be placed before 'from .base import BaseSerializer' (wrong-import-order)
apps/api/plane/app/serializers/page.py:15:0: C0411: third party import 'from plane.db.models import Page, PageLabel, Label, ProjectPage, Project, PageVersion' should be placed before 'from .base import BaseSerializer' (wrong-import-order)

Your code has been rated at 7.75/10 (previous run: 7.75/10, +0.00)
```

(Example of code quality score for file before modification with potential issues)

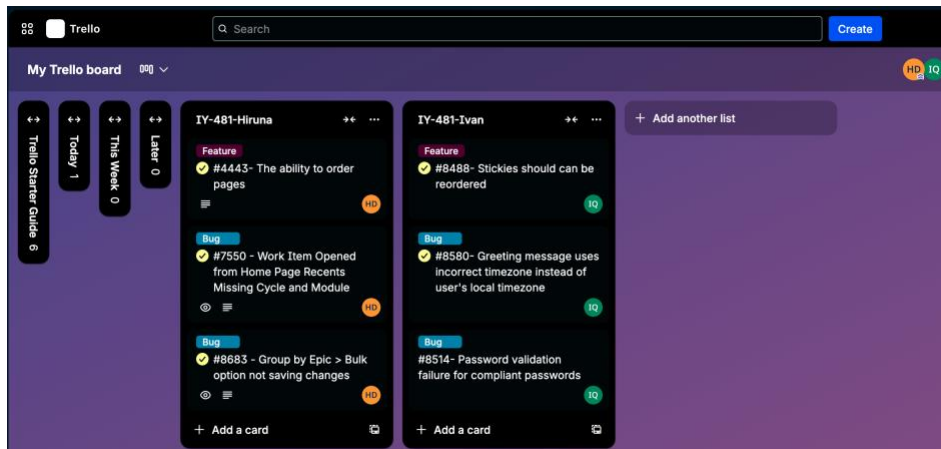
4. Maintainability Metrics: These metrics were collected using Radon to help us understand the complexity of modules that were being affected by changes.

```
(base) user@Yevins-MacBook-Pro IV-481-Plane % radon mi apps/api/plane/app/serializers/
apps/api/plane/app/serializers/intake.py - A
apps/api/plane/app/serializers/cycle.py - A
apps/api/plane/app/serializers/user.py - A
apps/api/plane/app/serializers/importer.py - A
apps/api/plane/app/serializers/asset.py - A
apps/api/plane/app/serializers/webhook.py - A
apps/api/plane/app/serializers/notification.py - A
apps/api/plane/app/serializers/_init_.py - A
apps/api/plane/app/serializers/page.py - A
apps/api/plane/app/serializers/view.py - A
apps/api/plane/app/serializers/favorite.py - A
apps/api/plane/app/serializers/estimate.py - A
apps/api/plane/app/serializers/module.py - A
apps/api/plane/app/serializers/workspace.py - A
apps/api/plane/app/serializers/issue.py - A
apps/api/plane/app/serializers/analytic.py - A
apps/api/plane/app/serializers/draft.py - A
apps/api/plane/app/serializers/base.py - A
apps/api/plane/app/serializers/project.py - A
apps/api/plane/app/serializers/state.py - A
(base) user@Yevins-MacBook-Pro IV-481-Plane %

(base) user@Yevins-MacBook-Pro IV-481-Plane % radon cc apps/api/plane/app/serializers/
apps/api/plane/app/serializers/intake.py
C 27:0 IntakeIssueSerializer - A
M 40:4 IntakeIssueSerializer.validate - A
M 48:4 IntakeIssueSerializer.update - A
C 73:0 IntakeIssueDetailSerializer - A
M 118:4 IntakeIssueDetailSerializer.to_representation - A
M 86:4 IntakeIssueSerializer.to_representation - A
C 27:0 IntakeSerializer - A
C 120:0 IntakeIssueLiteSerializer - A
C 127:0 IssueStateIntakeSerializer - A
apps/api/plane/app/serializers/cycle.py
C 15:0 CycleWriteSerializer - B
M 16:4 CycleWriteSerializer.validate - B
C 46:0 CycleSerializer - A
C 92:0 CycleIssueSerializer - A
C 162:0 CycleUserPropertiesSerializer - A
apps/api/plane/app/serializers/user.py
C 98:0 UserMetaSettingsSerializer - B
M 94:4 UserMetaSettingsSerializer.get_workspace - B
C 173:0 ChangePasswordSerializer - A
C 15:0 UserSerializer - A
M 131:4 ChangePasswordSerializer.validate - A
M 16:4 UserSerializer.validate_first_name - A
M 21:4 UserSerializer.validate_last_name - A
C 63:0 UserMetaSerializer - A
C 141:0 UserLiteSerializer - A
C 166:4 UserAdminLiteSerializer - A
C 193:0 ResetPasswordSerializer - A
C 281:0 ProfileSerializer - A
C 288:0 AccountSerializer - A
apps/api/plane/app/serializers/importer.py
C 13:0 ImporterSerializer - A
apps/api/plane/app/serializers/asset.py
C 9:0 FileAssetSerializer - A
apps/api/plane/app/serializers/webhook.py
C 19:0 WebhookSerializer - C
M 57:4 WebhookSerializer.update - C
M 22:4 WebhookSerializer.create - C
C 98:0 WebhookLogSerializer - A
```

(Maintainability index of files and Cyclomatic complexity of methods in files i.e the number of possible execution paths in a given file)

5. Assign issues to team members: Once we determined change requests that needed to be implemented our team used Trello to assign work to each team member ensuring work was tracked and not repeated unknowingly.



(Our teams Trello board listing activities and assignees to each task)

Bug Localization

1. This first involved using the tools in pre factoring activities to locate exactly where in the code base did we find the issues being listed.
2. Then we identified the failing code or behavior so that we could verify that the issue was correct. This was especially useful for back-end API points that interacted with the front end.
3. Then we traced the front-end interactions to these APIs allowing us to replicate issues or make UI changes as needed.

Change impact analysis (CIA)

After our team located modules that required modification, we performed CIA to gauge what components of the system might be affected by our modifications. The goal was to ensure that there would be no unintended side effects once the changes were implemented.

1. Back-end API logic

These modules handled request validation, sorting behavior and interactions between the front end and back end. Therefore, changes to these could affect how functionality such as pages are created, updated or retrieved. The following modules examined for impact analysis.

- apps/api/plane/app/serializers/page.py
- apps/api/plane/app/views/page/base.py
- apps/api/plane/api/views/sticky.py

2. Front end interface

These components were also analyzed due to their direct tie-in to the back end.

- apps/web/core/components/user-greetings.tsx

Implementation and refactoring

After completing CIA, we implemented the required changes.

Implementations Included

- Adding sorting to back end
- Updating front end to support drag page ordering
- Tracking page order

Refactoring included:

- Improve function structure
- Modularize code
- Maintain consistency with existing code
- Ensure functionality integrates cleanly with existing codebase

Testing

Our team implemented a testing layer verifying the changes functioned as expected such as page sorting and locking correctly. By including these tests using pytest we mitigated unintended side effects that could be caused by the modifications.

1. Unit test

- Unit tests were implemented to verify that new functionality behaved correctly
- For apps/api/plane/tests/contract/app/test_pages_app.py test cases, see “Files Altered” by Hiruna Yevin Dissanayake on page 2 of this document.

2. Regression Testing

- After unit tests were implemented, we ensured that our modifications did not break the existing functionality of the application
- Execute regression testing using this command
- `docker compose -f docker-compose-local.yml exec api python -m pytest plane/tests/contract/app/test_pages_app.py -q`

Code Review

Code review was conducted using the GitHub workflow, where all modifications were viewed through commits before being integrated into the repository.

Our team did the following:

- Verify correctness of implementation
- Ensure adherence to project coding standards
- Ensure sufficient test coverage
- Validate implementation fixed concerns in change request

Tools Used

- Team management: Trello
- Communication: Discord
- Back end: Docker
- Version Control: Git
- IDE: VS Code
- Code Search: ripgrep
- Code Analysis: Pylint
- Maintainability Metrics: Radon
- Testing: Pytest